PHYSICS-EL-01

Topics Covered: *Properties of charges; Charging by contact; Charging by induction Attraction and repulsion between charges*

Total Count of Questions: *25*

Expected Time of Attempt by Students: *50 min*

1. Which of the following is not a property of electric charge?
(a) Charges are additive
(b) Charges are conserved
(c) Charges exist only in multiples of 2e
(d) Like charges repel each other

2. Which statement is true about electric charges?
(a) They can be created or destroyed
(b) Only negative charges can move
(c) Electric charge is continuous
(d) Total charge of an isolated system remains constant

3. Which particle carries a negative charge?
(a) Proton (b) Neutron
(c) Electron (d) Positron

4. When a glass rod is rubbed with silk, it becomes positively charged because:
(a) It gains protons (b) It loses protons
(c) It gains electrons (d) It loses electrons

5. Which material is least likely to conduct electric charge?
(a) Copper (b) Aluminum
(c) Rubber (d) Silver

6. What happens when two like charges are brought near each other?
(a) They attract (b) They repel
(c) They merge (d) They neutralize

7. A charged object attracts a neutral object because of:
(a) Magnetic force
(b) Gravitational force
(c) Electrostatic induction
(d) Conduction

8. Which of the following is a method of charging a body?
(a) Heating
(b) Magnetic induction
(c) Conduction
(d) Vibration

9. In charging by contact, the final charges are:
(a) Opposite
(b) Equal and opposite
(c) Unequal
(d) Same sign

10. A body is said to be neutral when it has:
(a) More protons than electrons
(b) More electrons than protons
(c) Equal number of protons and electrons
(d) No protons or electrons

11. What is the SI unit of electric charge?
(a) Ampere (b) Coulomb
(c) Ohm (d) Volt

12. When a charged rod is brought near bits of paper, they get attracted. This happens due to:
(a) Repulsion
(b) Gravitational pull
(c) Induced polarization
(d) Magnetic attraction

13. A negatively charged object is repelled by another object. The second object is likely to be:
(a) Uncharged
(b) Negatively charged
(c) Positively charged
(d) Either positively charged or uncharged

14. What happens to the net charge in a closed system during any process?
(a) It increases (b) It decreases
(c) It remains constant (d) It becomes zero

15. Which of the following statements is correct?
(a) Only charged bodies exert force
(b) Uncharged bodies can exert electrostatic force
(c) Neutral bodies cannot interact with charges
(d) Charged bodies can attract or repel neutral bodies

16. What type of charge is acquired by a rubber rod when rubbed with fur?
(a) Positive (b) Negative
(c) Neutral (d) Double positive

17. Which process does not require physical contact?
(a) Charging by friction
(b) Charging by contact
(c) Charging by induction
(d) Discharging

18. When a charged body touches a neutral body, the neutral body gets charged by:
(a) Friction (b) Induction
(c) Contact (d) Radiation

19. Electric charge is:
(a) Continuous (b) Quantized
(c) Random (d) Imaginary

20. Which of the following supports the fact that electric charge is quantized?
(a) It flows continuously
(b) It is always positive
(c) It occurs in discrete units
(d) It has infinite values

21. If an object is grounded, what happens to excess charge on it?
(a) It gets amplified
(b) It disappears
(c) It gets transferred to Earth
(d) It remains on the object

22. A charged body can attract a neutral body due to:
(a) Repulsion
(b) Coulomb repulsion
(c) Force of conduction
(d) Induced charges

23. A conductor differs from an insulator in that a conductor:
(a) Has tightly bound electrons
(b) Has free electrons
(c) Cannot be charged
(d) Has no electrons

24. When an object is charged by induction, the charge it gets is:
(a) The same as the inducing charge
(b) Opposite to the inducing charge
(c) Neutral
(d) Variable

25. Two bodies with opposite charges are brought close to each other. They will:
(a) Repel (b) Attract
(c) Discharge each other (d) Disappear

ANSWERS

1. (c)	**2. (d)**	**3. (c)**	**4. (d)**	**5. (c)**	**6. (b)**	**7. (c)**	**8. (c)**	**9. (d)**	**10. (c)**
11. (b)	**12. (c)**	**13. (b)**	**14. (c)**	**15. (d)**	**16. (b)**	**17. (c)**	**18. (c)**	**19. (b)**	**20. (c)**
21. (c)	**22. (d)**	**23. (b)**	**24. (b)**	**25. (b)**					